- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

10 A boat pulls a water skier of mass 65 kg with a light inextensible horizontal towrope. The mass of the boat is 985 kg. There is a driving force of 2400 N acting on the boat. There are horizontal resistances to motion of 400 N and 1200 N acting on the skier and the boat respectively.

(a) Draw a diagram showing all the horizontal forces acting on the skier and the boat. **[2]**

(b) (i) Write down the equation of motion of the skier. **[1]**

(ii) Find the equation of motion of the boat. **[2]**

(c) Find the acceleration of the skier and the boat. **[1]**

The driving force of the boat is increased. The skier can only hold on to the towrope when the tension is no greater than her weight.

(d) Determine her greatest acceleration, assuming that the resistances to motion stay the same. **[2]**